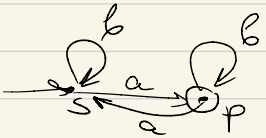
24.02

Крайни детерминирани автомати



$s \xrightarrow{\varepsilon} s \xrightarrow{\varepsilon} s$

$s \xrightarrow{b} s \xrightarrow{b} s \xrightarrow{a} p \xrightarrow{b} p \xrightarrow{a} s \xrightarrow{a} p$

$\mathcal{A} = \langle \Sigma, Q, s, \delta, F \rangle$

Σ – азбука (кр. мн-во)

Q – крайно мн-во

$\delta: Q \times \Sigma \to Q$ (ф-ция на преходите)

$F \subseteq Q$ (финални сост.)

$w = w_1 \dots w_n$ – дума над Σ

$p \in Q$

$p \xrightarrow{w_1} \delta(p, w_1) = p_1 \xrightarrow{w_2} \delta(p_1, w_2) = p_2 \xrightarrow{w_3} \dots \xrightarrow{w_n} p_n$

нарисаме път от p до p_n с етикет w в $\mathcal{A}$.

Език на $\mathcal{A}$ нарисаме:

$\mathcal{L}(\mathcal{A}) = \{ w \in \Sigma^* \mid$ от s има път с w до някое фин. сост. $f \}$

$= \{ w \in \Sigma^* \mid \exists f \in F \ (s \xrightarrow{w}_{\mathcal{A}}{}^{*} f) \}$

$\longmapsto \equiv \longrightarrow^*$

Задача: Нека $L = \{w \in \{0,1\}^* \mid w$ има четен брой $0\}$. Да се построи DFA с език L

Реш: За дума $w \in \{0,1\}^*$, $|w|_0$ - брой нули в w

Тогава $L = \{w \in \{0,1\}^* \mid |w|_0 \equiv 0 \pmod 2\}$

q_0 - началко сост.; q_0 "обещава да разпознае" L

фик. сост.	0	1	език	ЕЕ?
$\to * q_0$	q_1	q_0	$L = L_0$	да
q_1	q_0	q_1	$L' = \{w' \in \{0,1\}^* \mid \|w'\|_0 \equiv 1(2)\}$	не

$L = \{w \in \{0,1\}^* \mid |w|_0 \equiv 0(2)\}$

q_0: Какво ще стане, ако първата буква на w е 0?

$L = \{w' \mid 0w' \in L\} = \{w' \mid |0w'|_0 \equiv 0(2)\} = \{w' \mid 1 + |w|_0 \equiv 0(2)\}$
$= \{w' \mid |w'|_0 = 1(2)\}$

q_0: Какво ще стане, ако първата буква на w е 1?

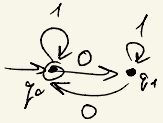
$L'' = \{w'' \mid 1w'' \in L\} = \{w'' \mid |1w''|_0 \equiv 0(2)\} = \{w'' \mid |w''|_0 \equiv 0(2)\}$

q_1: ——— // ——— (като q_0 за 0)

$L_1' = \{w' \mid 0w' \in L'\} = \{w' \mid |0w'| \equiv 1(2)\} = \{w' \mid 1 + |w'|_0 \equiv 1(2)\}$
$= \{w' \mid |w'|_0 \equiv 0(2)\} = L = L_0$

q_1: — // — 1?

$L_2' = \{w' \mid 1w' \in L'\} = \{w' \mid |1w'|_0 \equiv 1(2)\} = \{w' \mid |w'|_0 \equiv 1(2)\} = L'$



$|\varepsilon|_0 = 0 \equiv 0 \pmod 2$

Деф: За език $L \subseteq \Sigma^*$, $w \in \Sigma^*$ $\quad w^{-1}L \overset{def}{=} \{v \in \Sigma^* \mid wv \in L\}$

$w^{-1} \rightsquigarrow$ матрица
$L \rightsquigarrow$ вектор
$w^{-1}L \rightsquigarrow$ вектор

$L' = \{w' \in \{0,1\}^* \mid 0w' \in L\} = 0^{-1}L$
$L' = 0^{-1}L$

$L'' = \{w' \in \{0,1\}^* \mid 1w' \in L\} = 1^{-1}L$
$0^{-1}L' = L$
$1^{-1}L' = L'$

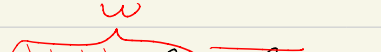
	0	1	език
→* q_0	q_1	q_0	$L = L_0 = 1^{-1}L$
q_1	q_0	q_1	$0^{-1}L_0 = L_1$

Св-ва: 1) $w \in L \iff \varepsilon \in w^{-1}L$

Ако $w = w\varepsilon \in L$, то ε е едно от тези v-та, за които $wv \in L$, т.е. $\varepsilon \in w^{-1}L$

Ако $\varepsilon \in w^{-1}L$, то ε е една от тези v-та, за които $v \in w^{-1}L$, т.е. $wv \in L \Rightarrow w\varepsilon \in L$
w

2) w, u - думи; L е език
$(wu)^{-1}L = u^{-1}(w^{-1}L)$



Тв: Нека Σ е азбука, $L = L_0, L_1, \ldots, L_n$ са езици над Σ, а $q_0 \ldots q_n$ са състояния на някакъв дет. авт. $A = (\Sigma, Q, q_0, \delta, F)$ със св-ва:

1). $q_i \in F \iff \varepsilon \in L_i$.

2). За всеки $i, j \le n$, $a \in \Sigma$: $\delta(q_i, a) = q_j \iff a^{-1}L_i = L_j$.

Тогава:

1) За всяка $w \in \Sigma^*$ и вс. i: $q_0 \xrightarrow{w}{}^* q_i \iff L_i = w^{-1} L_0$

2) $\mathcal{L}(A) = L_0$

Задача: Да се построи краен дет. авт с език:

$L = \{ w \in \{a,b\}^* \mid w \text{ съдържа } aba \}$

	a	b	език	$\varepsilon \in ?$
$\to q_0$	q_1	q_0	$L_0 = \{a,b\}^* \{aba\} \{a,b\}^*$	не
q_1	q_1	q_2	$L_1 = L_0 \cup \{ba\}\{a,b\}^*$	не
q_2	q_3	q_0	$L_2 = L_0 \cup \{a\}\{a,b\}^*$	не
$*q_3$	q_3	q_3	$L_3 = \{a,b\}^*$	да

1) За q_0, L_0:

$a^{-1} L_0 = a^{-1} (\{a,b\}^* \{aba\} \{a,b\}^*)$

$= a^{-1} ((\{a,b\}\{a,b\}^* \cup \{\varepsilon\}) \circ \{aba\}\{a,b\}^*)$

$= a^{-1} (\{a,b\}\{a,b\}^* \{aba\}\{a,b\}^* \cup \{aba\}\{a,b\}^*)$

$= a^{-1}\{a,b\}\{a,b\}^*\{aba\}\{a,b\}^* \cup a^{-1}\{aba\}\{a,b\}^*$

$a^{-1}\{a\}\{a,b\}^*\{aba\}\{a,b\}^*$

$\cup$

$a^{-1}\{b\}\{a,b\}^*\{aba\}\{a,b\}^* = \emptyset$

$\Rightarrow = \{a,b\}^*\{aba\}\{a,b\}^* \cup \{ba\}\{a,b\}^*$

$= L_0 \cup \{ba\}\{a,b\}^*$

$b^{-1}L_0 = b^{-1}(\{a,b\}^*\{aba\}\{a,b\}^*) =$

$= b^{-1}(\{a,b\}\{a,b\}^*\{aba\}\{a,b\}^* \cup \{aba\}\{a,b\}^*)$

$= b^{-1}\{a,b\}\{a,b\}^*\{aba\}\{a,b\} \cup \underbrace{b^{-1}\{aba\}\{a,b\}^*}_{\emptyset}$

$= \{a,b\}^*\{aba\}\{a,b\}^* = L_0$

2) За q_1, L_1:

$a^{-1}L_1 = a^{-1}(L_0 \cup \{ba\}\{a,b\}^*) = a^{-1}L_0 \cup \underbrace{a^{-1}\{ba\}\{a,b\}^*}_{\emptyset} =$

$= L_1 \cup \emptyset = L_1$

$b^{-1}L_1 = b^{-1}(L_0 \cup \{ba\}\{a,b\}^*) = b^{-1}L_0 \cup b^{-1}\{ba\}\{a,b\}^* = L_0 \cup \{a\}\{a,b\}^*$

$= L_2$

3) За q_2, L_2:

$a^{-1}L_2 = a^{-1}(L_0 \cup \{a\}\{a,b\}^*) = a^{-1}L_0 \cup a^{-1}\{a\}\{a,b\}^* = L_1 \cup \{a,b\}^*$

$= \{a,b\}^*$

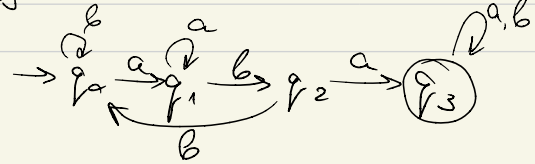
$b^{-1}L_2 = b^{-1}(L_0 \cup \{a\}\{a,b\}^*) = b^{-1}L_1 \cup \underbrace{b^{-1}\{a\}\{a,b\}^*}_{\emptyset} = L_2$

4) За q_3, L_3:

$a^{-1}L_3 = a^{-1}\{a,b\}^* = a^{-1}(\{a,b\}\{a,b\}^* \cup \{\varepsilon\})$

$= \underline{a^{-1}\{a,b\}}\{a,b\}^* \cup \underbrace{a^{-1}\{\varepsilon\}}_{\emptyset} = \{a,b\}^*$

$b^{-1}L_3 \overset{\text{аналог.}}{=} \{a,b\}^*$

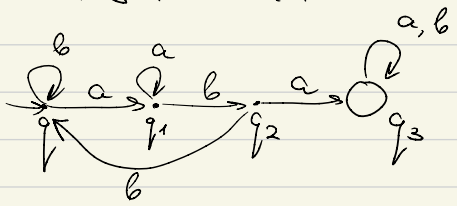


10.03

$\mathcal{L} \subseteq \{a,b\}^*$

$a^{-1}\mathcal{L} = \{\alpha \mid a\alpha \in \mathcal{L}\}$
$b^{-1}\mathcal{L} = \{\alpha \mid b\alpha \in \mathcal{L}\}$

$\varepsilon \in \mathcal{L}$?
$\mathcal{L} = \{a,b\}^* \{aba\} \{a,b\}^*$



	a	b
q_0	q_1	q_0
q_1	q_1	q_2
q_2	q_3	q_0
q_3	q_3	q_3

$\omega = \overset{1}{b}\overset{2}{b}\overset{3}{a}\overset{4}{b}\overset{5}{a}\overset{6}{a}\overset{7}{b}$

$p_0 = q_0$
$p_1 = \delta(p_0, b) = q_0$
$p_2 = \delta(p_1, b) = q_0$
$p_3 = \delta(p_2, a) = q_1$
$p_4 = \delta(p_3, b) = q_2$
$p_5 = \delta(p_4, a) = q_3$
$p_6 = \delta(p_5, a) = q_3$
$p_7 = \delta(p_6, b) = q_3$

Тъй като $q_3 \in F$, то $\omega \in \mathcal{L}(A)$

```
p ← p0
for i ← 1 to |w| do
    p ← δ(p, w[i])
return p ∈ F
```

$\alpha \in \{0,1\}^+$

$\bar{\alpha} = \sum_{i=0}^{n-1} 2^{n-1-i} \alpha_i$, $\alpha = \alpha_0 \alpha_1 \dots \alpha_{n-1}$, $\alpha_i \in \{0,1\}$

$\alpha = 00101$

$\bar{\alpha} = 5$

Зад: Да се построи КДА с език:

$L = \{\alpha \in \{0,1\}^+ \mid \bar{\alpha} \equiv 1 \pmod 5 \text{ или } \bar{\alpha} \equiv 2 \pmod 5\}$

$\overline{111} = 7 \equiv 2 \pmod 5 \Rightarrow \in L$

$\overline{00111} = \overline{111}$

$\overline{0\alpha} = \bar{\alpha}$

$\overline{1\alpha} = 2^{|\alpha|} + \bar{\alpha}$ $\quad \bar{\varepsilon} = 0$

Метод на Бързовски за $L = L_0$

	0	1	език	$\varepsilon \in$?	
→q_0	q_0	q_1	$L_0 = \{\alpha \mid \bar{\alpha} \equiv 1,2 \pmod 5\}$	не	
*q_1	q_2	q_3	$L_1 = \{\alpha \mid 2^{\|\alpha\|} + \bar{\alpha} \equiv 1,2 \pmod 5\}$	да	$(2^0 + 0 = 1 \equiv 1 \pmod 5)$
*q_2	q_4	q_0	$L_2 = \{\alpha \mid 2 \cdot 2^{\|\alpha\|} + \bar{\alpha} \equiv 1,2 \pmod 5\}$	да	
q_3	q_1	q_2	$L_3 = \{\alpha \mid 3 \cdot 2^{\|\alpha\|} + \bar{\alpha} \equiv 1,2 \dots\}$	не	
q_4	q_3	q_4	$L_4 = \{\alpha \mid 4 \cdot 2^{\|\alpha\|} + \bar{\alpha} \equiv \dots\}$	не	

1) $q_0: 0^{-1}L_0 = \{\alpha \mid 0\alpha \in L_0\} = \{\alpha \mid \overline{0\alpha} \equiv 1,2 \pmod 5\} = \{\alpha \mid \overline{\alpha} \equiv 1,2 \pmod 5\} = L_0$

$1^{-1}L_0 = \{\alpha \mid 1\alpha \in L_0\} = \{\alpha \mid \overline{1\alpha} \equiv 1,2 \pmod 5\} = \{\alpha \mid 2^{|\alpha|} + \overline{\alpha} \equiv 1,2 \pmod 5\} = L_1$

2) $q_1: 0^{-1}L_1 = \{\alpha \mid 0\alpha \in L_1\} = \{\alpha \mid 2^{|0\alpha|+1} + \overline{0\alpha} \equiv 1,2 \pmod 5\} =$ ($|0\alpha| = |\alpha|$, $\overline{0\alpha} = \overline{\alpha}$)

$= \{\alpha \mid 2 \cdot 2^{|\alpha|} + \overline{\alpha} \equiv 1,2 \pmod 5\} = L_2$

$1^{-1}L_1 = \{\alpha \mid 1\alpha \in L_1\} = \{\alpha \mid 2^{|1\alpha|} + \overline{1\alpha} \equiv 1,2 (\dots)\} = \{\alpha \mid 2 \cdot 2^{|\alpha|} + 2^{|\alpha|} + \overline{\alpha} \equiv 1,2 (\dots)\}$

$= \{\alpha \mid 3 \cdot 2^{|\alpha|} + |\alpha| \equiv 1,2 (\dots)\} = L_3$

3) $q_2: 0^{-1}L_2 = \{\alpha \mid 0\alpha \in L_2\} = \{\alpha \mid 2 \cdot 2^{|0\alpha|+1} + \overline{0\alpha} \equiv 1,2 (\dots)\} = \{\alpha \mid 4 \cdot 2^{|\alpha|} + \overline{\alpha} = \dots\} = L_4$ ($|0\alpha| = |\alpha|$, $\overline{0\alpha} = \overline{\alpha}$)

$1^{-1}L_2 = \{\alpha \mid 1\alpha \in L_2\} = \{\alpha \mid 2 \cdot 2^{|\alpha|+1} + 2^{|\alpha|} + \overline{\alpha} \equiv \dots\} = \{\alpha \mid \underbrace{5 \cdot 2^{|\alpha|}}_{\text{kratko na } 5} + \overline{\alpha} \equiv \dots\} =$

$= \{\alpha \mid \overline{\alpha} \equiv 1,2 (\dots)\} = L_0$

4) $q_3: 0^{-1}L_3 = \{\alpha \mid 0\alpha \in L_3\} = \{\alpha \mid 3 \cdot 2^{|\alpha|+1} + \overline{\alpha} \equiv \dots\} = \{\alpha \mid 6 \cdot 2^{|\alpha|} + \overline{\alpha} \equiv \dots\} =$

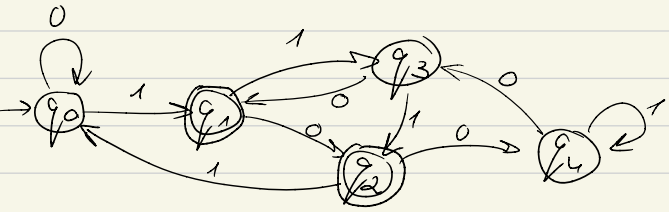
$= \{\alpha \mid 2^{|\alpha|} + \overline{\alpha} \equiv \dots\} = L_1$ ($6 \equiv 1 \pmod 5$)

$1^{-1}L_3 = \{\alpha \mid 1\alpha \in L_3\} = \{\alpha \mid 6 \cdot 2^{|\alpha|} + 2^{|\alpha|} + \overline{\alpha} \equiv \dots\} = \{\alpha \mid 7 \cdot 2^{|\alpha|} + \overline{\alpha} \equiv \dots\} = L_2$ ($7 \equiv 2 \pmod 5$)

5) $q_4: 0^{-1}L_4 = \{\alpha \mid 0\alpha \in L_4\} = \{\alpha \mid 8 \cdot 2^{|\alpha|} + \overline{\alpha} \equiv \dots\} = L_3$ ($8 \equiv 3 \pmod 5$)

$1^{-1}L_4 = \{\alpha \mid 1\alpha \in L_4\} = \{\alpha \mid 8 \cdot 2^{|\alpha|} + 2^{|\alpha|} + \overline{\alpha} \equiv \dots\} = \{\alpha \mid 9 \cdot 2^{|\alpha|} + \overline{\alpha} \equiv \dots\}$

$= L_4$ ($9 \equiv 4 \pmod 5$)



ДР: $\overline{\alpha} = \sum_{i=0}^{n-1} 5^{n-1-i} \alpha_i$ $\quad \alpha = \alpha_0 \dots \alpha_{n-1}$

$L = \{\alpha \in \{1,3,4\}^+ \mid 2\overline{\alpha} \equiv 3 \pmod 7\}$

Да се постр. КДА.

$\Sigma = \{\begin{bmatrix}0\\0\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix}, \begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}1\\1\end{bmatrix}\}$

За $\alpha \in \Sigma^*$: $\alpha = \begin{bmatrix}a_1\\b_1\end{bmatrix}\begin{bmatrix}a_2\\b_2\end{bmatrix} \dots \begin{bmatrix}a_n\\b_n\end{bmatrix}$, $\alpha^{(1)} = a_1 \dots a_n$, $\alpha^{(2)} = b_1 \dots b_n$

$y = *x + *$

$y = 2x + 4$

$L = \{\alpha \in \Sigma^* \mid \overline{\alpha^{(2)}} = 2 \cdot \overline{\alpha^{(1)}} + 4\}$

Нека $L' = \{\alpha^{rev} \mid \alpha \in L\}$

$\alpha = \alpha_0 \alpha_1 \dots \alpha_n$

$\alpha^{rev} = \alpha_n \dots \alpha_0$

Зад: Да се построи КДА с език

$L' = \{\alpha^{rev} \mid \alpha \in L\} = \{\alpha \in \Sigma^* \mid \overline{\alpha^{(2)rev}} = 2 \cdot \overline{\alpha^{(1)rev}} + 4\}$

$\alpha = \begin{bmatrix}a_0\\b_0\end{bmatrix} \dots \begin{bmatrix}a_{n-1}\\b_{n-1}\end{bmatrix}$

$\overline{\alpha^{(1)rev}} = a_0 + 2a_1 + 2^2 a_2 + \dots + 2^{n-1} a_{n-1} = \sum_{i=0}^{n-1} 2^i a_i$

$\overline{\alpha^{(2)rev}} = b_0 + 2b_1 + 2^2 b_2 + \dots + 2^{n-1} b_{n-1} = \sum_{i=0}^{n-1} 2^i b_i$

? $2\alpha^{\overline{(1)\,rev}} + 4 = \alpha^{\overline{(2)\,rev}}$

$2(a_0 + 2a_1 + \dots + 2^{n-1}a_{n-1}) + 4 = b_0 + 2b_1 + \dots + 2^{n-1}b_{n-1}$

$b_0 \in \{0, 1, 4\}$

От лявата страна имаме четно число, от дясната b_0 + четно.
Сл. $b_0 (\in \{0,1,4\})$ е 0.

Тъй като $2b_1 + \dots + 2^{n-1}b_{n-1}$ е четно и
$2a_1 + \dots + 2^{n-1}a_{n-1}$ е четно, а $b_0 \in \{0,1,4\}$, то
$b_0 = *.a_0 + *\pmod 2$
В нашия сл.
$b_0 = 2.a_0 + 4 \pmod 2$, т.е. $0 = b_0$
След като $b_0 = 0$, делим остатъчната част на 2:

$2(a_0 + 2a_1 + \dots) + 4 = b_0 + \dots + 2^{n-1}.b_{n-1} \;|:2$
$a_0 + 2a_1 + \dots + 2^{n-1}a_{n-1} + 2 = b_1 + \dots + 2^{n-2}b_{n-1}$
$2(a_1 + 2a_2 + 2^2a_3 + \dots + 2^{n-2}a_{n-1}) + a_0 + 2 = b_1 + \dots + 2^{n-2}b_{n-1}$

$\alpha = \boxed{\begin{matrix} a_0 \\ b_0 \end{matrix}}\beta$, то $2.\beta^{\overline{(1)\,rev}} + a_0 + 2 = \beta^{\overline{(2)\,rev}}$

6	17
00110	10001

6: 0 0 1 1 0 $\xrightarrow{rev}$ $\boxed{0}$ 1 1 0 0
17: 1 0 0 0 ① $\boxed{①}$ 0 0 0 1

$1 \neq 2.0 + 4 \pmod 2$

$2.6 + 4 \stackrel{?}{=} 17$
$16 \stackrel{?}{=} 17$
↓ ↓
четно нечет.

6 16

00110 10000

0 0 1 1 0 rev → 0 1 1 0 0

1 0 0 0 0 → 0 0 0 0 1

$b_0 = 2.0 + 4 \pmod 2$

$16_? \stackrel{?}{=} 2.6 + 4 \;/:2$

$8 \stackrel{?}{=} 2.3 + 2$

$a_0 = 0$

$a_0 + 2 = 2$

Вътр. дали $\begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \in L'$ е еквив. на вътр. дали

$\begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ изп. дет. ?

17.03.

$\alpha = a_0 \dots a_{n-1} \in \{0,1\}^*$

$\tilde{\alpha} = a_0 + 2a_1 + \dots + 2^{n-1} a_{n-1} = \sum_{i=0}^{n-1} 2^i a_i$

Тогава:

$\widetilde{0\alpha} = 0 + 2\tilde{\alpha}$

$\widetilde{1\alpha} = 1 + 2\tilde{\alpha}$

Интересуват ни $\alpha, \beta \in \{0,1\}^*$:

$\tilde{\beta} = 2\tilde{\alpha} + 4$

$\tilde{0}\beta = 2.\tilde{0}\alpha + 4 \Rightarrow 2\tilde{\beta} = 2.2.\tilde{\alpha} + 4 \;/:2 \Rightarrow \tilde{\beta} = 2\tilde{\alpha} + 2$

$\tilde{1}\beta = 2.\tilde{\alpha} + 4 \Rightarrow 1 + 2\tilde{\beta} = 2\tilde{\alpha} + 4$. Няма такива α и β

$\tilde{0}\beta = 2.\tilde{1}\alpha + 4 \Rightarrow 2\tilde{\beta} = 2 + 2.2\tilde{\alpha} + 4 \;/:2 \Rightarrow \tilde{\beta} = 2\tilde{\alpha} + 3$

Зад:

$\Sigma = \{\begin{bmatrix}0\\0\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix}, \begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}1\\1\end{bmatrix}\}$. Да се постр. КДА с език

$L = \{\alpha \in \Sigma^* \mid \tilde{\alpha}^{(2)} = 2.\tilde{\alpha}^{(1)} + 4\}$, където, ако

$\alpha = \begin{bmatrix}a_0\\b_0\end{bmatrix}\begin{bmatrix}a_1\\b_1\end{bmatrix}\cdots\begin{bmatrix}a_{n-1}\\b_{n-1}\end{bmatrix}$, $\alpha^{(1)} = a_0a_1\ldots a_{n-1}$, $\alpha^{(2)} = b_0b_1\ldots b_{n-1}$

	$\begin{bmatrix}0\\0\end{bmatrix}$	$\begin{bmatrix}0\\1\end{bmatrix}$	$\begin{bmatrix}1\\0\end{bmatrix}$	$\begin{bmatrix}1\\1\end{bmatrix}$	език	$\varepsilon \in$?
$\to q_0$	q_1				$L_0 = L = \{\alpha \mid \tilde{\alpha}^{(2)} = 2.\tilde{\alpha}^{(1)} + 4\}$	
q_1					$L_1 = \{\alpha \mid \tilde{\alpha}^{(2)} = 2\tilde{\alpha}^{(1)} + 2\}$	

Реш: q_0, L_0: $\begin{bmatrix}0\\0\end{bmatrix}^{-1}L_0 = \{\alpha \mid \tilde{0\alpha}^{(2)} = 2.\tilde{0\alpha}^{(1)} + 4\} = \{\alpha \mid \tilde{2.\alpha}^{(2)} = 2^2.\tilde{\alpha}^{(1)} + 4^2\} =$

$= \{\alpha \mid \tilde{\alpha}^{(2)} = 2\tilde{\alpha}^{(1)} + 2\} = L_1$

Дълго и задно, сл:

Нека $\alpha^{(i)} = \{\alpha \in \Sigma^* \mid \tilde{\alpha}^{(2)} = 2.\tilde{\alpha}^{(1)} + i\}$, $i \in \mathbb{N}$. Интересува ни $i = 4$

Нека $i \geq 0$

$\emptyset = \begin{bmatrix}0\\0\end{bmatrix}^{-1}L^{(i)} = \{\alpha \mid \tilde{0\alpha}^{(2)} = 2.\tilde{0\alpha}^{(1)} + i\} = \{\alpha \mid 2.\tilde{\alpha}^{(2)} = 2.2.\tilde{\alpha}^{(1)} + i\} = \begin{cases} \emptyset, & i \equiv 1 (2) \\ \{\alpha \mid \tilde{\alpha}^{(2)} = 2\tilde{\alpha}^{(1)} + \frac{i}{2}\}, & i \equiv 0 (2) \end{cases}$

$L_{\frac{i}{2}}$, $\frac{i}{2} \in \mathbb{N}$

$\mathcal{L}^{((i-1)/2)}$, $\frac{i-1}{2} \in \mathbb{N}$ или $\emptyset$

$\boxed{0\,1}^{-1}\mathcal{L}^{(i)} = \{\alpha \mid 1\tilde{\alpha}^{(2)} = 2\cdot 0\tilde{\alpha}^{(1)} + i\} = \{\alpha \mid 1 + 2\tilde{\alpha}^{(2)} = 2\cdot 2\cdot\tilde{\alpha}^{(1)} + i\}$

$= \{\alpha \mid 2\tilde{\alpha}^{(2)} = 4\cdot\tilde{\alpha}^{(1)} + i - 1\} = \begin{cases} \emptyset, & i \equiv 0(2) \\ \{\alpha \mid \tilde{\alpha}^{(2)} = 2\cdot\tilde{\alpha}^{(1)} + \frac{i-1}{2}\}, & i \equiv 1(2) \end{cases}$

$\boxed{1\,0}^{-1}\mathcal{L}^{(i)} = \{\alpha \mid 0\tilde{\alpha}^{(2)} = 2\cdot 1\tilde{\alpha}^{(1)} + i\} = \{\alpha \mid 2\tilde{\alpha}^{(2)} = 2(1 + 2\tilde{\alpha}^{(1)}) + i\} =$

$= \{\alpha \mid 2\tilde{\alpha}^{(2)} = 4\cdot\tilde{\alpha}^{(1)} + i + 2\} = \begin{cases} \emptyset, & i \equiv 1(2) \\ \{\alpha \mid \tilde{\alpha}^{(2)} = 2\tilde{\alpha}^{(1)} + \frac{i+2}{2}\}, & i \equiv 0(2) \end{cases}$

$\mathcal{L}^{((i+2)/2)}$

$\boxed{1\,1}^{-1}\mathcal{L}^{(i)} = \{\alpha \mid 1\tilde{\alpha}^{(2)} = 2\cdot 1\tilde{\alpha}^{(1)} + i\} = \{\alpha \mid 1 + 2\tilde{\alpha}^{(2)} = 2\cdot(1 + 2\cdot\tilde{\alpha}^{(1)}) + i\} =$

$= \{\alpha \mid 2\tilde{\alpha}^{(2)} = 4\tilde{\alpha}^{(1)} + i + 1\} = \begin{cases} \emptyset, & i \equiv 0(2) \\ \mathcal{L}^{((i+1)/2)}, & i \equiv 1(2) \end{cases}$

$i: \frac{i+2}{2} > i$ $\quad i < 2$ $\quad i = 0, 1$ $\qquad i: \frac{i+1}{2} > i$ $\quad i < 1$ $\quad i = 0$

Пусть $Q = \{q_0, q_1, q_2, q_3, q_4, q_\emptyset\}$, $\delta(q_i, \boxed{0\,0}) = \begin{cases} q_{i/2}, & i \equiv 0(2) \\ q_\emptyset, & i \equiv 1(2) \end{cases}$

$\delta(q_i, \boxed{0\,1}) = \begin{cases} q_{\frac{i-1}{2}}, & i \equiv 1(2) \\ q_\emptyset, & i \equiv 0(2) \end{cases}$ $\qquad \delta(q_i, \boxed{1\,0}) = \begin{cases} q_{i/2+1}, & i \equiv 0(2) \\ q_\emptyset, & i \equiv 1(2) \end{cases}$

$\delta(q_i, \boxed{1\,1}) = \begin{cases} q_{\frac{i+1}{2}}, & i \equiv 1(2) \\ q_\emptyset, & i \equiv 0(2) \end{cases}$

Тъй като за $0 \le i \le 4$, $\frac{i}{2} + 1 \le 2 + 1 \le 4$, $\frac{i-1}{2} \ge 0$ за $i \equiv 1(2)$,

$\frac{i-1}{2} \le i \le 4$, то δ е добре деф. $\delta : (Q \times \Sigma) \to \Sigma$

Тъй като:

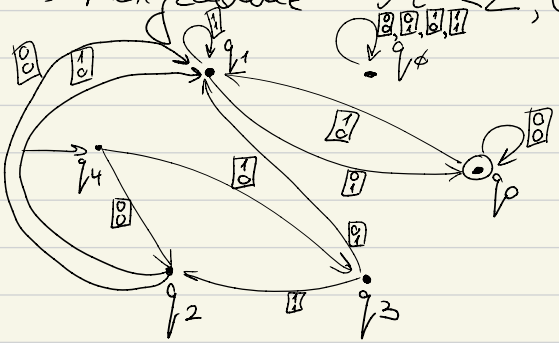
$\mathcal{L} = \mathcal{L}^{(4)}$ избираме кат. сост. q_4.

Кои $\mathcal{L}^{(i)}$ съдържат ε?

$\varepsilon \in \mathcal{L}^{(i)} \iff \tilde{\varepsilon}^{(2)} = 2 \cdot \tilde{\varepsilon}^{(1)} + i \iff 0 = 2 \cdot 0 + i \iff i = 0$

$\Rightarrow$ фин. сост. ще бъде точно q_0

$\Rightarrow$ Получаваме $A = \langle \Sigma, Q, q_4, \delta, \{q_0\} \rangle$



Зад. Да се постр. КДА с език

$\mathcal{L} = \{\alpha \in \Sigma^* \mid 2\tilde{\alpha}^{(2)} \ge 3\tilde{\alpha}^{(1)} + 2\}$

$\Sigma, \alpha^{(2)}, \alpha^{(1)}$ са свързите думи

$2 \cdot 0\tilde{\alpha} \ge 3\,0\tilde{\beta} + 2 \iff 2 \cdot 2 \cdot \tilde{\alpha} \ge 3 \cdot 2 \cdot \tilde{\beta} + 2 \overset{/:2}{\iff} 2\tilde{\alpha} \ge 3\tilde{\beta} + 1$

$2 \cdot 1\tilde{\alpha} \ge 3 \cdot 1\tilde{\beta} + 2 \iff 2 + 4\tilde{\alpha} \ge 3 + 6\tilde{\beta} + 2 \overset{/:2}{\iff} 2\tilde{\alpha} \ge 3\tilde{\beta} + \frac{3}{2}$

Само те се пренасят

Нека $\mathcal{L}^{(i)} = \{\alpha \in \Sigma^* \mid 2.\tilde{\alpha}^{(2)} \geq 3.\tilde{\alpha}^{(1)} + i\}$, $i \in \mathbb{Z}$

$2x \geq 3y + \frac{3}{2} \overset{x,y \in \mathbb{Z}}{\iff} 2x \geq 3y + \frac{3+1}{2}$, защото най-малкото цяло число $\geq \frac{3}{2}$ е 2 и щом $2x$ и $3y$ са цели, $2x-3y$ е цяло, т.е. $2x-3y \geq \frac{3}{2}$ вле-зе $2x-3y \geq 2$

$$[0\,0]\,\mathcal{L}^{(i)} = \{\alpha \mid 2.0\tilde{\alpha}^{(2)} \geq 3.0\tilde{\alpha}^{(1)} + i\} = \{\alpha \mid 4.\tilde{\alpha}^{(2)} \geq 6.\tilde{\alpha}^{(1)} + i\} = \{\alpha \mid 2\tilde{\alpha}^{(2)} \geq 3.\tilde{\alpha}^{(1)} + \tfrac{i}{2}\}$$

$$\overset{\text{ЛЕМА}}{=} \begin{cases} \{\alpha \mid 2\tilde{\alpha}^{(2)} \geq 3\tilde{\alpha}^{(1)} + \frac{i}{2}\}, & i \equiv 0(2) \\ \{\alpha \mid 2\tilde{\alpha}^{(2)} \geq 3\tilde{\alpha}^{(1)} + \frac{i+1}{2}\}, & i \equiv 1(2) \end{cases}$$

$$= \begin{cases} \mathcal{L}^{(i/2)}, & i \equiv 0(2) \\ \mathcal{L}^{((i+1)/2)}, & i \equiv 1(2) \end{cases} = \mathcal{L}^{(\lceil \frac{i}{2} \rceil)}$$

$$[0\,1]\,\mathcal{L}^{(i)} = \{\alpha \mid 2.1\tilde{\alpha}^{(2)} \geq 3.0\tilde{\alpha}^{(1)} + i\} = \{\alpha \mid 2 + 4\tilde{\alpha}^{(2)} \geq 6.\tilde{\alpha}^{(1)} + i\} = \{\alpha \mid 2\tilde{\alpha}^{(2)} \geq 3\tilde{\alpha}^{(1)} + \tfrac{i-2}{2}\}$$

$$= \begin{cases} \mathcal{L}^{(\frac{i}{2}-1)}, & i \equiv 0\,(2) \\ \mathcal{L}^{(\frac{i+1}{2}-1)}, & i \equiv \,(2) \end{cases}$$

$$[1\,0]\,\mathcal{L}^{(i)} = \{\alpha \mid 2.0\tilde{\alpha}^{(2)} \geq 3.1\tilde{\alpha}^{(1)} + i\} = \{\alpha \mid 4.\tilde{\alpha}^{(2)} \geq 3 + 6\tilde{\alpha}^{(1)} + i\} = \{\alpha \mid 2\tilde{\alpha}^{(2)} \geq 3\tilde{\alpha}^{(1)} + 1 + \tfrac{i+1}{2}\}$$

$$= \begin{cases} \mathcal{L}^{(1+\frac{i+1}{2})}, & i \equiv 1(2) \\ \mathcal{L}^{(\frac{i}{2}+2)}, & i \equiv 0(2) \end{cases}$$

$$[1\,1]\,\mathcal{L}^{(i)} = \{\alpha \mid 2.1\tilde{\alpha}^{(2)} \geq 3.1\tilde{\alpha}^{(1)} + i\} = \{\alpha \mid 2 + 4\tilde{\alpha}^{(2)} \geq 3 + 6\tilde{\alpha}^{(1)} + i\} = \{\alpha \mid 2\tilde{\alpha}^{(2)} \geq 3\tilde{\alpha}^{(1)} + \tfrac{i+1}{2}\}$$

$$= \begin{cases} \mathcal{L}^{(\frac{i+1}{2})}, & i \equiv 1(2) \\ \mathcal{L}^{(\frac{i}{2}+1)}, & i \equiv 0(2) \end{cases}$$

Интересува ни $\mathcal{L} = \mathcal{L}^{(2)}$

$\frac{i}{2}-1, \frac{i}{2}, \frac{i+1}{2}, 1+\frac{i+1}{2}, \frac{i}{2}+2, \frac{i}{2}+1$

$|\frac{i}{2} - const| < |i|$ за i достатъчно голямо

Искаме $|\frac{i}{2}+2| \leq |i|$

$|\frac{i}{2}-1| \leq |i|$

[0 0]	[0 1]	[1 0]	[1 1]
q_{2k}	q_k	q_{k-1}	q_{k+2}	q_{k+1}
q_{2k+1}	q_{k+1}	q_k	q_{k+2}	q_{k+1}

Началко сост.: $q_2 \rightsquigarrow \mathcal{L}^{(2)} = \{\alpha | 2\alpha^{(2)} \geq 3\alpha^{(1)} + 2\}$

Финални сост.: $q_i \rightsquigarrow \varepsilon \in \mathcal{L}^{(i)} = \{\alpha | 2\alpha^{(2)} \geq 3\alpha^{(1)} + i\} \Leftrightarrow 2\tilde{\varepsilon} \geq 3\tilde{\varepsilon} + i \Leftrightarrow 2.0 \geq 3.0 + i$

$\boxed{i \leq 0}$

Кои са състоянията?

$\frac{i}{2}+2 \leq \frac{c}{2}$, ако $|i| \leq c$

$\frac{i}{2}-1 \geq -c$, ако $|i| \leq c$

т.е. $\frac{c}{2}+2 \leq c \quad c \geq 4$

$-\frac{c}{2}-1 \geq -c \quad , c \geq 2$

Т.е ако $Q = \{q_k | k \geq -4, k \leq 4\}$ с δ, как. сост. q_2 и фин. $\{q_{-4}, q_{-3}, q_{-2}, q_{-1}, q_0\}$ е КДА за A

24.03.

TODO

31.03.

Детерминизация на КА О

Зад. Да се детерм. автоматът:

	0	1	2
→p	p	p,q	p
q	—	—	r
r	—	r,s	s
*s	q,r,s	—	—

Реш:

	0	1	2
→{p}	{p}	{p,q}	{p}
{p,q}	{p}	{p,q}	{p,r}
{p,r}	{p}	{p,q,r,s}	{p,s}
*{p,q,r,s}	{p,q,r,s}	{p,q,r,s}	{p,r,s}
*{p,s}	{p,q,r,s}	{p,q}	{p}
*{p,r,s}	{p,q,r,s}	{p,q,r,s}	{p,s}

ми-ва от нач. сост + тези до които достигнем от тях с ε от когато

прежди: P – множ. от сост;
α – буква

1) P_α – сост. до които с преход α от Φ може да стигнем до тях

2) $\delta(\Phi, \alpha)$ – множ. от сост, до които може да стигнем от P_α с ε.

Финални: Онези ми-ва, които съдържат фин. сост.

Зад: Да се детерминизира авт:

	a	b
→0	0,2,3	0
→1	—	1,3
2	2,3	2,4
*3	—	4
*4	—	—

Реш:

	a	b
→{0,1}	{0,2,3}	{0,1,3}
*{0,2,3}	{0,2,3}	{0,2,4}
*{0,1,3}	{0,2,3}	{0,1,3,4}
*{0,2,4}	{0,2,3}	{0,2,4}
*{0,1,3,4}	{0,2,3}	{0,1,3,4}

Зад: —//— :

	a	b	ε
*1	—	4	3
→2	1	—	1
3	1	—	2
4	7	1	—
*5	6	—	—
*6	9	1	—
7	6	7	5
8	6	—	9
9	—	7	8

Реш:

	a	b
*→{1,2,3}	{1,2,3}	{4}
{4}	{5,7}	{1,2,3}
*{5,7}	{6}	{5,7}
*{6}	{8,9}	{1,2,3}
{8,9}	{6}	{5,7}

Зад: —//—

	a	b	c	ε
→1	–	–	–	2
2	3	–	–	1
3	–	–	–	4
4	–	5	–	6
5	–	–	8	–
6	–	7	–	3
7	4	–	–	–
8	–	–	–	9
*9	–	–	–	8,1
*→10	–	–	–	–

Реш:

	a	b	c
*→{1,2,10}	{3,4,6}	∅	∅
{3,4,6}	∅	{5,7}	∅
∅	∅	∅	∅
{5,7}	{3,4,6}	∅	{1,2,8,9}
*{1,2,8,9}	{3,4,6}	∅	∅

Минимизация на КДА

Дадено: ДКА $A = <\Sigma, Q, s, \delta, F>$

Търси се: $A_м = <\Sigma, Q_м, s_м, \delta_м, F_м>$

1) $A_м$ тот. КДА с $\mathcal{L}(A_м) = \mathcal{L}(A)$

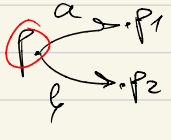
2) $|Q_M|$ е възм. най-малко изм/у $\forall$ авт. със св-во 1)

I) Премахване недостижими сост. от $\subseteq$ в A

II) Ако получеяният авт. не е тот., го попълваме

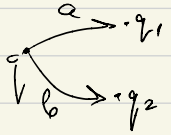
III) $reg \longmapsto A_p = <\Sigma, Q, p, \delta, F>$

$p \sim q \overset{def}{\Leftrightarrow} \mathcal{L}(A_p) = \mathcal{L}(A_q)$



$\mathcal{L}(A_p) = \{a\}\mathcal{L}(A_{p_1}) \cup \{b\}\mathcal{L}(A_{p_2}) \cup \{\varepsilon\}$

? || || ? || ? ||



$\mathcal{L}(A_q) = \{a\}\mathcal{L}(A_{q_2}) \cup \{b\}\mathcal{L}(A_{q_2})$

Зад: Да се намери ДСТКДА еквив. на:

	a	b
→0	1	2
1	3	4
2	5	3
3	6	7
4	7	6
5	7	7
*6	1	1
*7	2	1

Рем:

I. Достижимост! BFS
В нашия сл. $\forall$ са достижими

II. Тотален е авт!

III. (Вместо $p \overset{?}{\sim} q$, итеративно намираме кл. на еквив. на рел. $p \sim^{(i)} q \Leftrightarrow \mathcal{L}(A_p) \cap \Sigma^{\leq i} = \mathcal{L}(A_q) \cap \Sigma^{\leq i}$ докато $\sim^{(i)} = \sim^{(i+1)}$. Тогава $\sim^{(i)} = \sim$!)

1) Разбиваме сост. на финални и нефинални

$F, Q \setminus F$ - кл. на еквив. на $\sim^{(0)}$

		a	b
0	→0	1	2
1	1	3	4
1	2	5	3
2	3	6	7
2	4	7	6
2	5	7	7
3	*6	1	1
3	*7	2	1

$B_0 = \{0,1,2,3,4,5\}$ - нефин.
$B_1 = \{6,7\}$ - фин.

$\sim^{(0)} \longrightarrow \sim^{(1)}$

Разгл. B_0 и B_1:

B_0	a	b
0	0	0
1	0	0
2	0	0
3	1	1
4	1	1
5	1	1

$B_0^{(1)} = \{0,1,2\}$, $B_1^{(1)} = \{3,4,5\}$, $B_2^{(1)} = \{6,7\}$

B_1	a	b
6	0	0
7	0	0

$\sim^{(0)} \neq \sim^{(1)}$
$\sim^{(1)} \longrightarrow \sim^{(2)}$
Разгл. $B_0^{(1)}, B_1^{(1)}, B_2^{(1)}$

$B_0^{(1)}$	a	b
0	0	0
1	1	1
2	1	1

$B_1^{(1)}$	a	b
3	2	2
4	2	2
5	2	2

$B_2^{(1)}$	a	b
6	0	0
7	0	0

$B_0^{(2)} = \{0\}$ $B_2^{(2)} = \{3,4,5\}$

$B_1^{(2)} = \{1,2\}$ $B_3^{(2)} = \{6,7\}$

$B_a^{(2)}$ – синглетон (няма да се разбие!)

$B_1^{(2)}$	a	b
1	2	2
2	2	2

стабилен

$B_2^{(2)}$	a	b
3	3	3
4	3	3
5	3	3

стабилен

$B_3^{(2)}$	a	b
6	1	1
7	1	1

стабилен

Сл. $\sim^{(2)} = \sim^{(3)}$ и сл. $\sim^{(2)} = \sim$

Окончателно:

	a	b
→0	1	1
1	2	2
2	3	3
*3	1	1

кл. на еквив

Преходите идват от табл.

Нач. сост – клас, който съдържа нач. сост. от оригиналния авт

Фин. – кл., които съдържат финални

07.04.

Зад: Да се минимизира КДА:

	a	b
✓. →1	2	3
✓. 2	1	7
✓. *3	5	1
– 4	9	5
✓. 5	7	8
– 6	3	5
✓. *7	8	2
✓. 8	3	5
–*9	6	1

Рем: I. Достижимост от нач. сост.
4,6,9 не са!

		a	b
1 1	→1	2	3
1 1	2	1	7
0 0	*3	5	1
2 1	5	7	8
0 0	*7	8	2
2 1	8	3	5

II. Попълваемако не е тотален. Тотален е (няма недеф. прехо-ди)

III. Тест на 1-та буква

1) $B_0 = \{3,7\}$ - финални, $B_1 = \{1,2,5,8\}$ - нефинални

B_0	a	b
3	1	1
7	1	1

стабилен

B_1	a	b	
1	1	0	+
2	1	0	+
5	0	1	–
8	0	1	–

$\sim^{(0)} \longrightarrow \sim^{(1)}$

$B_0^{(1)} = \{3, 7\}$, $B_1^{(1)} = \{1, 2\}$, $B_2^{(1)} = \{5, 8\}$

$B_0^{(1)}$	a	b
3	2	1
7	2	1

стабилен

$B_1^{(1)}$	a	b
1	1	0
2	1	0

стабилен

$B_2^{(1)}$	a	b
5	0	2
8	0	2

стабилен

$\Rightarrow \sim^{(1)} = \sim^{(2)}$ сл. процеса завършва, $\sim^{(2)} = \sim$

	a	b
*0	2	1
→1	1	0
2	0	2

е МТКДА екв. на дадения

Т-ма (Myhill-Nerode): $\mathcal{L} \subseteq \Sigma^*$ е рег. $\Leftrightarrow$ има краен брой разл. $w^{-1}\mathcal{L}$ за $w \in \Sigma^*$

$(u \equiv_{\mathcal{L}} v \overset{def}{\Leftrightarrow} u^{-1}\mathcal{L} = v^{-1}\mathcal{L})$

Зад. Да се докаже, че $\{a^{n^2} \mid n \in \mathbb{N}\}$ не е рег.

Д-во: Нека $\mathcal{L} = \{a^{n^2} \mid n \in \mathbb{N}\}$

Нека $n < m$. $n^2 + 2n + 1 = (n+1)^2 \Rightarrow a^{n^2+2n+1} = a^{(n+1)^2} \in \mathcal{L}$. Сл. $a^{2n+1} \in (a^{n^2})^{-1}\mathcal{L}$. Сега, ако $(a^{n^2})^{-1}\mathcal{L} = (a^{m^2})^{-1}\mathcal{L}$, то $a^{2n+1} \in (a^{m^2})^{-1}\mathcal{L}$ и сл. $m^2 + 2n + 1 = k^2$ за някое $k \in \mathbb{N}$.

Но $m^2 + 2n + 1 > m^2$

$m^2 + 2n + 1 < m^2 + 2m + 1 = (m+1)^2$. Сл. $m^2 < k^2 < (m+1)^2$ и тъй като $k \in \mathbb{N}$, то това е невъзможно. Сл. $(a^{n^2})^{-1}\mathcal{L} \neq (a^{m^2})^{-1}\mathcal{L}$, т.е. $a^{n^2} \not\equiv_{\mathcal{L}} a^{m^2}$ за $n < m$

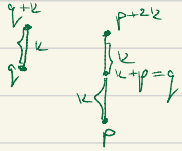
Сл. броят на кл. на екв. на $\equiv_{\mathcal{L}}$ е ∞. Сл. (Myhill-Nerode) $\mathcal{L}$ не е рег.

Зад: Да се докаже, че $\{a^p \mid p \text{ - просто}\}$ не е рег.

Реш: Нека $p<q$ - прости. Нека $k = q-p \geq 1$ и да докажем, че $a^p \equiv_{\mathcal{L}} a^q$ (т.е. $(a^p)^{-1}\mathcal{L} = (a^q)^{-1}\mathcal{L}$)

Тъй като $k = q-p$, то $p+k = q$ е просто. Сл. $a^k \in (a^p)^{-1}\mathcal{L} = (a^q)^{-1}\mathcal{L}$

Сл. $a^{q+k} \in \mathcal{L}$, т.е. $q+k$ е просто. Но $q+k = (p+k)+k = p+2k$



Сега (по инд.), ако $p+\ell k$ е просто, то $p+(\ell+1)k$ е просто, защото $p+\ell k$ е просто е еквив. на $a^{\ell k} \in (a^p)^{-1}\mathcal{L}$ и сл. $a^{q+\ell k} \in \mathcal{L}$, т.е. $q+\ell k$ е просто. Но $q+\ell k = (p+k)+\ell k = p+(\ell+1)k$.

Сл. за $\forall \ell \in \mathbb{N}$: $p+\ell k$ е просто.

В частност $\ell = p$, $p+pk$ е просто, т.е. $p(k+1)$ е просто. Но $k \geq 1$ и сл. $k+1 \geq 2$. Сл. $p(k+1)$ не е просто ↯

$\Rightarrow a^p \not\equiv_{\mathcal{L}} a^q$ за $p \neq q$. Тъй като простите числа са ∞ $\Rightarrow \equiv_{\mathcal{L}}$ е ∞

$\Rightarrow \mathcal{L}$ не е рег.

Зад: Да се докаже, че:

1) $\{a^n b^m \mid n \geq m,\ n,m \in \mathbb{N}\} = \mathcal{L}_1$ не е рег.
2) $\{a^n b^n \mid n \in \mathbb{N}\} = \mathcal{L}_2$ — // —
3) $\{w \in \{a,b\}^* \mid |w|_a = |w|_b\} = \mathcal{L}_3$ — // —

Реш:

1) Нека $n \neq m$, $n<m$. Разг. a^n и a^m. Да докажем, че $a^n \equiv_{\mathcal{L}_1} a^m$. Сл. от това, че $n \geq n$, $a^n b^n \in \mathcal{L}_1$ и сл. $b^n \in (a^n)^{-1}\mathcal{L}_1 = (a^m)^{-1}\mathcal{L}_1$
Сега $a^m b^n \in \mathcal{L}_1$ ✓

Обратно, $a^m b^m \in L_1$, сл. $b^m \in (a^m)^{-1} L_1 = (a^n)^{-1} L_1 \Rightarrow a^n b^m \in L_1$. Но $n < m$.

Сл. $a^n b^m \notin L_1$ $\Rightarrow a^n \equiv_{L_1} a^m$, $n \neq m$. Сл. $\text{ind}(\equiv_{L_1}) = \infty$

2) Да док-ме, че L_2 е рег. $L_2 = \{a^n b^n \mid n \in \mathbb{N}\}$. Ако сега разгледаме

$\{a^k \underbrace{a^n}_{\|} b^n \mid k, n \in \mathbb{N}\} = \{a\}^* L_2$

$\{a^{n+k} b^n \mid k, n \in \mathbb{N}\} = \{a^m b^n \mid m \geq n\} = L_1$. Сл. $L_1 = \{a\}^* \cdot L_2$ и ако L_2 беше рег., то и L_1 щеше да е. Но той не е! Сл. L_2 не е рег.

3) $L_3 \cap \{a\}^* \{b\}^* = \{a^n b^m \mid |a^n b^m|_a = |a^n b^m|_b\} = \{a^n b^m \mid n = m, n, m \in \mathbb{N}\}$
$= \{a^n b^n \mid n \in \mathbb{N}\} = L_2$

Ако L_3 беше рег., $\cap$ запазва регулярността ($\{a\}^* \{b\}^*$ - рег.)

$\Rightarrow$ и L_2 щеше да е рег., но той не е $\Rightarrow L_3$ не е рег.

Зад: Да се док-е, че $\{ww \mid w \in \{a, b\}^*\}$ не е рег.

Реш: Нека $n, m \in \mathbb{N}$ и да док-ме, че $a^n b \equiv_L a^m b$. Тогава

$a^n b a^n b \in L \Rightarrow a^n b \in (a^n b)^{-1} L = (a^m b)^{-1} L$. Сл. $a^m b a^n b \in L$, т.е.
$a^m b a^n b = ww$ $(w \in \{a, b\}^*)$

Сл. w завършва с b, w започва с a. Сл. тъй като има точно две b в $a^m b a^n b$, то $w = a^m b$, $w = a^n b$. Сл. $a^n b = a^m b$, т.е. $n = m$

$\Rightarrow a^n b \equiv_L a^m b \Leftrightarrow m = n$

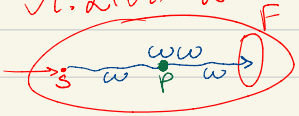
Сл. $\equiv_L$ има безбр. много кл. на еквив.

Сл. L не е рег.

Зад: Нека L е рег. език. Докажете, че $\{w \mid ww \in L\} = L'$ е рег.

регулярност → регулярни операции
→ дет. кр. авт.
→ кл. на екв. на $= L'$

$A: L(A) = L$ KDA



Реш: Нека $A = \langle \Sigma, G, s, \delta, F \rangle$ е KDA с език L
Тогава $w \in L' \Leftrightarrow ww \in L \Leftrightarrow s \xrightarrow{ww}{}^* F \Leftrightarrow$ има $p \in G$ ($s \xrightarrow{w}{}^* p$ и $p \xrightarrow{w}{}^* F$)

За състояние $p \in G$ може да разгледаме:

$\overrightarrow{A_p} = \langle \Sigma, G, p, \delta, F \rangle$

$\overleftarrow{A_p} = \langle \Sigma, G, s, \delta, \{p\} \rangle$

Сл. $w \in L' \Leftrightarrow$ има $p \in G$ ($s \xrightarrow{w}{}^* p$ и $p \xrightarrow{w}{}^* F$)
$\Leftrightarrow$ има $p \in G$ ($w \in L(\overleftarrow{A_p})$ и $w \in L(\overrightarrow{A_p})$)
$\Leftrightarrow$ има $p \in G$ ($w \in L(\overleftarrow{A_p}) \cap L(\overrightarrow{A_p})$)
$\Leftrightarrow w \in \bigcup_{p \in G} (L(\overleftarrow{A_p}) \cap L(\overrightarrow{A_p}))$

Сл. $L' = \bigcup_{p \in G} (L(\overleftarrow{A_p}) \cap L(\overrightarrow{A_p}))$

Тъй като $\overleftarrow{A_p}$ и $\overrightarrow{A_p}$ са KDA $\Rightarrow$ езиците им са рег. $\Rightarrow$ сеч. им също $\Rightarrow$ и обед. $\Rightarrow L'$ е рег.

Зад: Нека L е рег. език. Докажете, че $L' = \{w \mid \exists u (wu \in L \wedge |u| = |w|)\}$ е рег.

Реш:

$w \in L' \Leftrightarrow$ има u ($wu \in L$, $|u| = |w|$).

Да забележим, че ако $A = \langle \Sigma, Q, s, \delta, F \rangle$ е VDA и разгл.
$A_{all} = \langle \Sigma, Q, s, \Delta_{all}, F \rangle$ $\Delta_{all} = \{\langle p, \sigma, q \rangle \mid \sigma \in \Sigma, p, q \in Q$ и $\exists a (\delta(p,a) = q)\}$

Тогава $s \xrightarrow{v}_{A_{all}}^{*} F \Leftrightarrow$ има $a_1, a_2, \dots, a_{|v|} \in \Sigma : s \xrightarrow{a_1 a_2 \dots a_{|v|}}_{A}^{*} F$

$\Rightarrow L(A_{all}) = \{w \in \Sigma^* \mid \exists u \in L(A) (|u| = |w|)\}$

Сега $w \in L' \Leftrightarrow$ има $p \in Q$ и u ($s \xrightarrow{w}_{A}^{*} p$ и $p \xrightarrow{u}_{A}^{*} F$ и $|u| = |w|$), където $A = \langle \Sigma, Q, s, \delta, F \rangle$ е VDA с език L. Тогава $w \in L' \Leftrightarrow$ има $p \in Q$ $w \in L(\overleftarrow{A}_p)$ и $w \in L(\overrightarrow{A}_p)_{all}$))

$\Leftrightarrow w \in \bigcup_{p \in Q} (L(\overleftarrow{A}_p) \cap L(\overrightarrow{A}_p)_{all}))$

$L' = \bigcup_{p \in Q} (\underbrace{L(\overleftarrow{A}_p)}_{\text{рег}} \cap \underbrace{L(\overrightarrow{A}_p)_{all}}_{\text{рег.}})) \Rightarrow L'$ е рег.

Д.Р. Да се докаже, че $\{ww^{rev} \mid w \in \{a,b\}^*\}$ не е рег.

Ако L е рег, то $\{w \mid ww^{rev} \in L\}$ е рег.

L - рег, дсд, че $\{w \mid \exists u (|w| = |u|$ и $wu \in L\}$ е рег.

21.04.

Контекстносвоб. грам. и езици

правила
нетерминали - <мета понятие>
терминали
аксиома

Деф:

Контекстносвоб. гр. е $G=<N,\Sigma,P,S>$, където $N\cap\Sigma=\emptyset$ са азбуки
$P\subseteq_{fin} N\times(\Sigma\cup N)^*$ крайно мн-во
от $A\to\alpha$, $A\in N$, $\alpha\in(\Sigma\cup N)^*$
$S\in N$

Нека $G=<N,\Sigma,P,S>$ е к.св. гр. Ако $x,y\in(N\cup\Sigma)^*$, а
$A\to\alpha\in P$ е правило, пишем:

$xAy \Rightarrow_G x\alpha y$

за една стъпка от xAy може да изведем $x\alpha y$ (като приложим $A\to\alpha$)

Извод в $G=<\dots>$ кар. редица $v_0\Rightarrow_G v_1\Rightarrow_G\dots\Rightarrow_G v_n$, където
v_i изхожда за една от v_{i+1} за $i\le n$
n е кар. дължина на извода

$v_0\Rightarrow_G^{(n)} v_n$

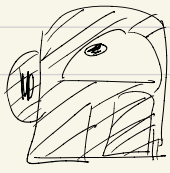
$v_0\Rightarrow^* v_n$, ако $\exists n: v_0\Rightarrow_G^{(n)} v_n$

Език на к.св. гр. $G=<\dots>$ кар. $\{w\in\Sigma^* \mid S\Rightarrow_G^* w\}=\mathcal{L}(G)$

Пр: $G=<\{S\},\{a,b\},\{S\to\varepsilon\mid aSb\},S>$

$S\Rightarrow^{(0)} S\notin\mathcal{L}(G)$ $\quad S\neq a,b$
$S\Rightarrow^{(1)} aSb\notin\mathcal{L}(G)$
$S\Rightarrow^{(1)}\varepsilon\in\mathcal{L}(G)$

$S\Rightarrow^{(1)} aSb\Rightarrow^{(1)} aaSbb\Rightarrow^{(1)} aaaSbbb\overset{S\to\varepsilon}{\Rightarrow} aaabbb\in\mathcal{L}(G)$



$G = \langle \{S, A, B\}, \{a, b\}, \{S \to AB \mid BA \mid SS,\ A \to a \mid AS,\ B \to b \mid BS\}, S \rangle$

$S \Rightarrow^{(1)} SS \overset{S \to AB}{\Rightarrow}^{(1)} ABS \Rightarrow ABBA \overset{A \to a}{\Rightarrow} ABBa \overset{B \to b}{\Rightarrow} ABba \overset{\substack{B \to a \\ A \to a}}{\Rightarrow} abba \in L(G)$

Задача за принадлежност:

$w \in \Sigma^*$

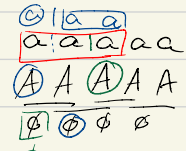
$G = \langle N, \Sigma, P, S \rangle$ с правила $\cdot A \to \gamma,\ \gamma \in \Sigma$

$\cdot A \to BC,\ B, C \in N$

Въпрос: $w \in L(G)$?

Да/Не

за горния пр:



$w = w_1 \dots w_n$

$V_{i,j} = \{ N \in N \mid N \Rightarrow^* w_i w_{i+1} \dots w_{i+j-1},\ j \le n - i + 1 \}$

Цел:

$V_{1,n} = ?$

$N \in V_{1,n} \iff N \Rightarrow w_1 w_2 \dots \underbrace{w_{1+n-1}}_{w_n}$

$w = aaaaa$

$V_{1,5} = \{ N \in \{A, S, B\} \mid N \Rightarrow^* w_1 \dots w_5 \} \to \{ N \in \{A, S, B\} \mid N \Rightarrow^* aaaaa \}$

$w \in L(G) \overset{def}{\iff} S \Rightarrow_G^* w$

От др. страна

$V_{i,1} = \{ N \in N \mid N \Rightarrow^* w_i \dots w_{i+1-1} \} = \{ N \in N \mid N \Rightarrow^* w_i \} = \{ N \in N \mid N \to w_i \in P \}$

$V_{i,j+1} = \{ N \in \mathcal{N} \mid N \to N'N'',\ N' \Rightarrow w_i \dots w_{i+k-1} \text{ и } N'' \Rightarrow w_{i+k} \dots w_{i+j} \}$
$N' \in V_{i,k},\ N'' \in V_{i+k,j-k+1}$

	a	a	a	a	a
i=1	A	A	A	A	
i=2	ϕ	ϕ	ϕ	ϕ	
i=3	ϕ	ϕ	ϕ		
i=4	ϕ	ϕ			
i=5	ϕ				

Зад: Да се провери дали $w = aabbba \in \mathcal{L}(G)$ (горното G)

	1 a	2 a	3 b	4 b	5 b	6 a
i=1	A	A	B	B	B	A
i=2	ϕ	S	ϕ	ϕ	S	
i=3	A	ϕ	ϕ	B		
i=4	S	ϕ	ϕ			
i=5	ϕ	ϕ				
i=6	S					

	1 a	2 a	3 b
1	A	A	B
2	ϕ	S	

A

$N \to AS$

a a b

a+ab U aa+b

aa+b

$\{A\} \cup \phi = \{A\}$

abb

ab+b a+bb

Сл. $S \in V_{1,6}$, т.е. $w = aabbba \in \mathcal{L}(G)$

Зад: $G = \langle \{S, A, B, C\}, \{a, b\}, \{S \to AB \mid BC,\ A \to BA \mid a,\ B \to CC \mid b,\ C \to AB \mid a\}, S \rangle$

Да се провери дали:
1) $baaba \in \mathcal{L}(G)$
2) $aaaaaa \in \mathcal{L}(G)$

1)

	1 b	2 a	3 a	4 b	5 a
1	B	A,C	A,C	B	A,C
2	A,S	B	S,C	A,S	
3	ϕ	B	B		
4	ϕ	A,S,C			

5 SC/A

$S \in V_{1,5} \Rightarrow baaba \in \mathcal{L}(G)$

$baab \stackrel{?}{\in} \mathcal{L}(G)$ - не

$aba \stackrel{?}{\in} \mathcal{L}(G)$ - не

28.04.

Контекстносвободни езици

К.св. език е $\mathcal{L} \subseteq \Sigma^*$, за който има к.св. гр. G: $\mathcal{L}(G) = \mathcal{L}$

Зад: Дк.д., че $\{a^n b^n \mid n \in \mathbb{N}\}$ е к.св.

Рем: Нека $G = \langle \{a,b\}, \{S\}, \{S \to \varepsilon \mid aSb\}, S \rangle$

Цел: $\mathcal{L}(G) = \{a^n b^n \mid n \in \mathbb{N}\}$

1). С инд. по n показваме, че $S \Rightarrow_G^* a^n b^n$

- $n=0$: В G има правило $S \to \varepsilon \Rightarrow S \Rightarrow_G \varepsilon = a^0 b^0$
- $n \to n+1$: Нека $S \Rightarrow_G^* a^n b^n$. Тъй като в G има правило $S \to aSb$, то $S \Rightarrow_G aSb \overset{\text{инд.}}{\Rightarrow}_G^* a(a^n b^n)b = a^{n+1} b^{n+1}$

2). Твърдим, че ако $S \Rightarrow_G^{(n)} w \in \{a,b\}^*$, то $w \in \{a^m b^m \mid m \in \mathbb{N}\}$

С инд. по n (пълна мат. инд.) показваме, че $(*)$ ако $S \Rightarrow^{(n)} w \in \{a,b\}^*$, то $w = a^m b^m$ за някое m.

Нека $n \in \mathbb{N}$ и за $n' < n$ $(*)$ е вярно за n'. Нека $S \Rightarrow^{(n)} w \in \{a,b\}^*$. Тъй като $w \in \{a,b\}^*$ и $S \Rightarrow^{(0)} S \notin \{a,b\}^*$, то $n > 0$. Сл. $S \Rightarrow \alpha \Rightarrow_G^{(n-1)} w$, където $S \to \alpha$ е правило.

1). $\alpha = \varepsilon$. Тогава $\varepsilon \Rightarrow_G^* \varepsilon$, т.е. $w = \varepsilon = a^0 b^0 \in \{a^m b^m \mid m \in \mathbb{N}\}$

2). $\alpha = aSb$. Тогава $aSb \Rightarrow_G^{(n-1)} w \in \{a,b\}^*$ и тъй като G е к.св.,

$G' = \langle \{a, b\}, \{S, s, f, S'\}, \{S' \to sS, s \to af, f \to af \mid \varepsilon\}, S' \rangle$

нов (S') — това — от G_{a^+} — от G: $S \to \varepsilon \mid aSb$

---

1). Ако $A = \langle \Sigma, Q, \{s\}, \Delta, F \rangle$ е краен. авт. и $Q \cap \Sigma = \emptyset$, то $G_A = \langle \Sigma, Q, P, s \rangle$, където $P = \{p \to aq \mid \langle p, a, q \rangle \in \Delta\} \cup \{f \to \varepsilon \mid f \in F\}$ има език $\mathcal{L}(G_A) = \mathcal{L}(A)$.

2). Ако $G_1 = \langle \Sigma, N_1, P_1, S_1 \rangle$ и $G_2 = \langle \Sigma, N_2, P_2, S_2 \rangle$ са к.св., $N_1 \cap N_2 = \emptyset$

2.1.) $G_\cup = \langle \Sigma, N_1 \cup N_2 \cup \{S\}, P_1 \cup P_2 \cup \{S \to S_1 \mid S_2\}, S \rangle$, където $S \notin N_1 \cup N_2$ има език $\mathcal{L}(G) = \mathcal{L}(G_1) \cup \mathcal{L}(G_2)$

2.2.) $G_\circ = \langle \Sigma, N_1 \cup N_2 \cup \{S\}, P_1 \cup P_2 \cup \{S \to S_1 S_2\}, S \rangle$, където $S \notin N_1 \cup N_2$ има език $\mathcal{L}(G) = -N- \circ -N-$

2.3.) $G_* = \langle \Sigma, N_1 \cup \{S\}, P_1 \cup \{S \to \varepsilon \mid S_1 S\}, S \rangle$, където $-N-$

Задача: ДСД, където

1) $L_1 = \{a^n b^m \mid n \neq m \in \mathbb{N}\}$

2) $L_2 = \{a^{n_1} b^{m_1} a^{n_2} b^{m_2} \dots a^{n_k} b^{m_k} \mid k \in \mathbb{N},\ n_i, m_i > 0,\ \forall 1 \le i \le k : n_i \neq m_i\}$

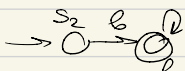
са к.св.

Реш:

1) $\{a\}^+ \circ \{a^n b^n\} \cup \{a^n b^n\} \circ \{b\}^+ = L_1 = \{a^n b^m \mid n > m \text{ или } m > n\}$

$G_{a^n b^n} = \langle \{a, b\}, \{S\}, \{S \to \varepsilon \mid aSb\}, S \rangle$

$G_{a^+} = \langle \{a, b\}, \{s_1, f_1\}, \{s_1 \to af_1, f_1 \to af_1 \mid \varepsilon\}, s_1 \rangle$



$G_{b^+} = \langle \{a, b\}, \{s_2, f_2\}, \{s_2 \to bf_2, f_2 \to bf_2 \mid \varepsilon\}, s_2 \rangle$

$G' = \langle \{a, b\}, \{S, s_1, f_1, S'\}, \{S' \to s_1 S_{f_1}, S'\}, S' \rangle = G_{a^+} \circ G_{a^n b^n}$

P_1: $s_1 \to af_1$, $f_1 \to af_1 \mid \varepsilon$; $S_{f_1} \to aSbf_1 \mid \varepsilon$

$G'' = \langle \{a,b\}, \{S_2, s_2, t_2, S''\}, \{S'' \to S_2 s_2, s_2 \to b t_2, t_2 \to b t_2 | \varepsilon, S_2 \to a S_2 b | \varepsilon\}, S'' \rangle$

P_2 ; $G'' = G_{a^n b^n} \cup G_{b^+}$

$G = \langle \{a,b\}, \{S_1, S_2, s_1, s_2, t_1, t_2, S', S'', S\}, \{S \to S' | S'' \cup P_1 \cup P_2, S\rangle$

$\mathcal{L}(G) = \mathcal{L}_1 \Rightarrow \mathcal{L}_1$ е к.св.

2) $\{a\}\mathcal{L}_1\{b\} = \{a\}\{a^n b^m | n \neq m \in \mathbb{N}\}\{b\} = \{a^{n+1} b^{m+1} | n \neq m\} = \{a^n b^m | n \neq m, n,m>0\}$

$\Rightarrow \mathcal{L}_2 = (\{a\}\mathcal{L}_1\{b\})^*$ - к.св.

05.05.

Зад: Да се построи к.св. гр. за $\mathcal{L} = \{a^n b^m c^k | m > \min(n,k)\}$

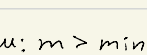
Реш: $m > \min(n,k)$ е еквив. на $m > n$ или $m > k$

$\Rightarrow \mathcal{L} = \{a^n b^m c^k | m > n \text{ или } m > k\} = \{a^n b^m c^k | m > n\} \cup \{a^n b^m c^k | m > k\}$

$= \{a^n b^m | m > n\} \cdot \{c\}^* \cup \{a\}^* \cdot \{b^m c^k | m > k\}$

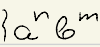
$= \{a^n b^n\}\{b\}^+\{c\}^* \cup \{a\}^+\{b\}^+\{b^k c^k\}$

$G_{a^n b^n} = \langle \{a,b,c\}, \{S_1\}, \{S_1 \to a S_1 b | \varepsilon\}, S \rangle$



$G_{b^+} = \langle \{a,b,c\}, \{s_1, t_1\}, \{s_1 \to b t_1, t_1 \to b t_1 | \varepsilon\}, s_1 \rangle$

$G_{c^*} = \langle \{a,b,c\}, \{s_2\}, \{s_2 \to c s_2 | \varepsilon\}, s_2 \rangle$



$G_{a^*} = \langle \{a,b,c\}, \{s_3\}, \{s_3 \to a s_3 | \varepsilon\}, s_3 \rangle$

$G_{b^k c^k} = \langle \{a,b,c\}, \{S_2\}, \{S_2 \to b S_2 c | \varepsilon\}, S_2 \rangle$

$G_{a^n b^n b^+} = \langle \{a,b,c\}, \{S_1, s_1', t_1', S_3\}, \{S_3 \to S_1 s_1', S_1 \to a S_1 b | \varepsilon, s_1' \to b t_1', t_1' \to b t_1' | \varepsilon\}, S_3 \rangle$

$G_{a^n b^n b^+ c^*} = \langle \{a,b,c\}, \{S_1, s_1', t_1', S_3, s_2, S_4\}, \{S_4 \to S_3 s_2, S_3 \to S_1 s_1', S_1 \to a S_1 b | \varepsilon, s_1' \to b t_1', t_1' \to b t_1' | \varepsilon, s_2 \to c s_2 | \varepsilon, S_4\rangle$

$G_{b^+b^{12}c^{12}} = \langle \{a,b,c\}, \{S_1'', S_2, S_5\}, \{S_5 \to S_1'' S_2,\ S_1'' \to b f_1'',\ f_1'' \to b f_1'' | \varepsilon\},\ S_5 \rangle$
$S_2 \to b S_2 c | \varepsilon$

$G_{a^* b^+ b^{12} c^{12}} = \langle \{a,b,c\}, \{S_3, S_1'', f_1'', S_2, S_5, S_6\}, \{S_6 \to S_3 S_5,\ S_3 \to a S_3 | \varepsilon,\ S_5 \to S_1'' S_2,\ S_1'' \to b f_1'',\ f_1'' \to b f_1'' | \varepsilon,\ S_2 \to b S_2 c | \varepsilon\}, S_6 \rangle$

$G_V = \langle \{a,b,c\}, \{S_2, S_1', f_1', S_4, S_3, S_4, S_3, S_1'', f_1'', S_2, S_5, S_6\}, \{S_7 \to S_6 | S_4,\ S_6 \to S_3 S_5,\ S_3 \to a S_3 | \varepsilon,\ S_5 \to S_1'' S_2,\ S_1'' \to b f_1',\ f_1' \to b f_1'' | \varepsilon,\ S_2 \to b S_2 c | \varepsilon,\ S_4 \to S_3 S_2,\ S_3 \to S_1 S_1',\ S_1 \to a S_1 b | \varepsilon,\ S_1' \to b f_1',\ f_1' \to b f_1' | \varepsilon,\ S_2 \to c S_2 | \varepsilon\}, S_7 \rangle$

Зад: ДСД, се изградите
1) $\mathcal{L}_1 = \{a^n b^m \mid |n-m| > 6\}$
2) $\mathcal{L}_2 = \{a^n b^m \mid |n-m| \le 6\}$
са к.сб.

Реш: 1) $\mathcal{L}_1 = \{a^n b^m \mid n-m < 6$ или $n-m < -6\} = \{a^n b^m \mid n > m+6\} \cup \{a^n b^m \mid m > n+6\}$
$= \{a\}^+ \{a^{6+m} b^m\} \cup \{a^n b^{n+6}\} \{b\}^+$
per. $\{a\}^6 \{a^m b^m\}$ к.сб. $\{a^n b^n\} \{b\}^6$ к.сб. per. per.

$\Rightarrow$ рег. $\circ$ к.сб. $=$ к.сб.
к.сб. $\cup$ к.сб. $=$ к.сб. $\Rightarrow \mathcal{L}_1$ е к.сб.

2) $|n-m| \le 6 \Leftrightarrow n-m \le 6$ и $n-m \ge -6 \Leftrightarrow n-m \in \{-6,-5,-4,\dots,4,5,6\} = \{i \in \mathbb{Z} \mid i \in [-6,6]\}$

$\mathcal{L}_2 = \bigcup_{i=-6}^{6} \{a^n b^m \mid n-m=i\} = \bigcup_{i=0}^{6} \{a^n b^m \mid n = m+i\} \cup \bigcup_{i=1}^{6} \{a^n b^m \mid \underset{m=n+i}{n-m=-i}\}$

$= \bigcup_{i=0}^{6} \{a^{m+i} b^m \mid m \in \mathbb{N}\} \cup \bigcup_{i=1}^{6} \{a^n b^{n+i} \mid n \in \mathbb{N}\} = \underset{\text{крайно}}{\bigcup_{i=0}^{6}} \{a^i\} \underset{\text{к.сб.}}{\{a^m b^m\}} \cup \underset{\text{к.сб.}}{\bigcup_{i=1}^{6}} \{a^n b^n\} \underset{\text{крайно}}{\{b^i\}}$

крайно крайно

$\Rightarrow L_2$ е к.сб.

Зад: $D \subset \Delta$, те ез.

1) $L_1 = \{a^n b^m \mid 3n > 2m\}$

2) $L_2 = \{a^n b^m \mid 2m \le n \le 5m\}$

са к.сб.

Реш: 1) $3n > 2m \Leftrightarrow n > \frac{2}{3} m$

$m = 3k, 3k+1, 3k+2, k \in \mathbb{N}$

Сботв при $m = 3k$: $n > \frac{2}{3} m \Leftrightarrow n > 2k$, $L_0' = \{a^n b^{3k} \mid n > 2k\}$

при $m = 3k+1$: $n > \frac{2(3k+1)}{3}$, т.е. $n > 2k + \frac{2}{3}$, $L_1' = \{a^n b^{3k+1} \mid n > 2k\}$

при $m = 3k+2$: $n > 2k + \frac{4}{3} = 2k+1+\frac{1}{3}$, $L_2' = \{a^n b^{3k+1} \mid n > 2k+1\}$

Сега $L_1 = L_0' \cup L_1' \cup L_2'$, ст. е дост. да докажем L_0', L_1', L_2' са к.сб

$L_0' = \{a^n b^{3k} \mid n > 2k\} = \{a^{2k+\ell} b^{3k} \mid k \in \mathbb{N}, \ell > 0\} = \{a\}^+ \{a^{2k} b^{3k} \mid k \in \mathbb{N}\}$

$L_1' = \{a^n b^{3k+1} \mid n > 2k\} = L_0' \circ b = \{a\}^+ \circ \{a^{2k} b^{3k} \mid k \in \mathbb{N}\} \circ \{b\}$

$L_2' = \{a^n b^{3k+2} \mid n > 2k+1\} = \{a^{2k+1+\ell} b^{3k+1} \mid k \in \mathbb{N}, \ell > 0\} \{b\} = \{a\} \circ L_1' \circ \{b\}$

По аналогия на $\{a^n b^n\}$ L_0' е к.сб. $\Rightarrow L_1', L_2'$ - к.сб. $\Rightarrow L_1$ е к.сб.

$(S \to aaSbbb \mid \varepsilon)$

12.05. 2) $a^2 b^2 \notin L_2$

$ab^3 \in L_2$

$a^2 b^4, \ldots, a^2 b^{10} \in L_2$

$2n = \underbrace{2 + 2 + \ldots + 2}_{n}$

$m = m_1 + m_2 + \ldots + m_n$ с $2 \le m_i \le 5$

$5n = \underbrace{5 + 5 + \ldots + 5}_{n}$

Нека $2n \le m \le 5n$. Тогава $m = m_1 + m_2 + \dots + m_n$ за някои $2 \le m_i \le 5$.

Тъй като $2n \le m \le 5n$, то $2n \le m < 3n$, $3n \le m < 4n$, $4n \le m < 5n$ или $m = 5n$.
Ако $m = 5n$, то може да изберем $m_i = 5$ за $i \le n$.
Иначе $kn \le m < (k+1)n$ и ако $m \equiv r \pmod n$.

$$m = \underbrace{kn + r}_{0 \le r < n} = \underbrace{k + k + \dots + k}_{n} + \underbrace{1 + \dots + 1}_{r} = r(k+1) + (n-r)k$$

Пр: $4.2 \le 10 < 4.5$

$4.2 \le 10 \le 4.3$

$10 \equiv 2 \pmod 4$

$$10 = \underbrace{2+2+2+2}_{4} + \underbrace{1+1}_{2 \pmod 4} = 3+3+2+2$$

Реш: Нека $G = \langle \{a, b\}, \{S\}, S \to \varepsilon \mid aSb^2 \mid aSb^3 \mid aSb^4 \mid aSb^5 \}, S \rangle$

Твърдим, че $\mathcal{L}(G) = \mathcal{L}_2$

1). С индукция по n:

$(*, n)$: $S \Rightarrow_G^{(n)} w \in \{a, b\}^*$, то $w = a^k b^m$ и $2k \le m \le 5k$

Док. $(*, n)$ с пълна мат. инд. по n:
Нека $n \in \mathbb{N}$ и за $\forall n' < n$: $(*, n')$ е вярно.

Нека $S \Rightarrow_G^{(n)} w \in \{a, b\}^*$. При $n = 0$, $w = S \notin \{a, b\}^* \Rightarrow n > 0$ и тогава:

$S \Rightarrow \alpha \Rightarrow_G^{(n-1)} w \in \{a, b\}^*$, където $S \to \alpha$ е правило в G

1сл. $S \to \alpha = S \to \varepsilon$, тогава $w = \varepsilon = a^0 b^0$ и тъй като $2.0 \le 0 \le 5.0$, то $\varepsilon \in \mathcal{L}_2$

2сл. $S \to \alpha = S \to aSb^\ell$, $\ell \in \{2, \dots, 5\}$. Тогава

$aSb^\ell = \alpha \Rightarrow_G^{(n-1)} w \in \{a,b\}^*$

С1. $w = aw'b^\ell$, където $S \Rightarrow_G^{(n-1)} w' \in \{a,b\}^*$. Ясно е, че $w' \in \{a,b\}^*$ (w' е поддума на $w \in \{a,b\}^*$)

От $(*, n-1) \Rightarrow w' \in L_2$, т.е. $w' = a^k b^m$, $2k \le m \le 5k$

$\Rightarrow w = aw'b^\ell = aa^k b^m b^\ell = a^{k+1} b^{m+\ell}$

и тъй като $2k \le m \le 5k$, $2 \le \ell \le 5$

$\Rightarrow 2k+2 \le m+\ell \le 5k+5$

$\Rightarrow w \in L_2$

Доказ $L(G) \subseteq L_2$.

Нека сега $a^n b^m \in L_2$, т.е. $2n \le m \le 5n$. Тогава $m = m_1 + \dots + m_n$ с $2 \le m_i \le 5$, $i \in \{1, \dots, n\}$

С1. тъй като за $2 \le m_i \le 5$ $S \to aSb^{m_i}$ е правило в G, то

$S \Rightarrow_G aSb^{m_1} \Rightarrow_G a^2 Sb^{m_1+m_2} \Rightarrow_G \dots \Rightarrow_G a^n Sb^{m_1+m_2+\dots+m_n} \Rightarrow_{S \to \varepsilon} a^n b^m$

т.е. $a^n b^m \in L(G) \Rightarrow L_2 \subseteq L(G)$

$\Rightarrow L_2 = L(G)$ и е к.св.

Зад: Да се построи к.св. грам. за $\mathcal{L} = \{a^n b^m c^{n+m} \mid n,m \in \mathbb{N}\}$

$$\begin{cases} S \to aSc \mid B \\ B \to bBc \mid \varepsilon \end{cases}$$

$a^n (b^m c^m) c^n$

$a a^n (b^m c^m) c^n c \quad (m,n) \to (m, n+1)$

$a^n (b^m bc c^m) c^n \quad (m,n) \to (m+1, n)$

$G = \langle \{a,b\}, \{S,T\}, \{T \to S \mid aTc, S \to \varepsilon \mid bSc\}, T \rangle$

Цел: $\mathcal{L}(G) = \{a^n b^m c^{n+m} \mid n,m \in \mathbb{N}\}$

Знаем,че $G' = \langle \{a,b,c\}, \{S\}, \{S \to \varepsilon \mid bSc\}, S \rangle$ има език $\mathcal{L}(G') = \{b^m c^m \mid m \in \mathbb{N}\}$

1) $\mathcal{L} \subseteq \mathcal{L}(G)$. С инд. по n ще $(*,n)$: $\forall m (a^n b^m c^{m+n} \in \mathcal{L}(G))$

• $n=0$. $T \to S$ е правило в G. Тъй като правилата на G' са правила и на G, то за $\forall m$ $S \Rightarrow_G^* b^m c^m$

Сл. $T \Rightarrow_G S \Rightarrow_G^* b^m c^m$ за $\forall m$.

• $n \mapsto n+1$. Нека $(*,n)$ е вярно. Разгл. произволно $m \in \mathbb{N}$

$a^{n+1} b^m c^{n+1+m} = a(a^n b^m c^{m+n}) c$

И.Х. $T \Rightarrow_G^* a^n b^m c^{m+n}$. Освен това $T \to aTc$ е правило в $G \Rightarrow T \Rightarrow_G aTc \Rightarrow_G^* a(a^n b^m c^{n+m})c = a^{n+1} b^m c^{n+1+m}$

2) Тъй като правилата в G с лява страна нетерм. от G' (S) са същите правилата на G', то ако $S \Rightarrow_{G'}^* w$, значи $S \Rightarrow_G^* w$

Тъй като $\mathcal{L}(G') = \{b^m c^m \mid m \in \mathbb{N}\}$, то $S \Rightarrow_G^* w \in \{a,b,c\}^*$, то $w \in \{b^m c^m \mid m \in \mathbb{N}\}$

С инд. по ℓ док-ем:

$(**, \ell): T \Rightarrow_G^{(\ell)} w \in \{a,b,c\}^*$ тогава $w \in \mathcal{L}$

Пълна мат. инд. по ℓ. Нека $(**, \ell')$ е вярно за $\forall \ell' < \ell$.

$T \Rightarrow_G^{(\ell)} w \in \{a,b,c\}^*$

$\ell \neq 0$ иначе $w = T \notin \{a,b,c\}^*$

Сл. $T \Rightarrow \alpha \Rightarrow_G^{(\ell-1)} w$, където $T \to \alpha$ е правило.

1 сл. $\alpha = S \Rightarrow S \Rightarrow_G^* w \in \{a,b,c\}^*$ и сл. $w = b^m c^m$ за някое m, т.е. $w = \underbrace{a^0 b^m c^m}_{\in \mathcal{L}}$

2 сл. $\alpha = aTc$. Тогава $aTc \Rightarrow_G^{(\ell-1)} w$, сл. $w = a w' c \in \{a,b,c\}^*$ и $T \Rightarrow_G^{(\ell-1)} w'$.

От $(**, \ell-1) \Rightarrow w' = a^n b^m c^{n+m}$ за някои n, m и сл. $w = a(\ldots)c \in \mathcal{L}$ ✓

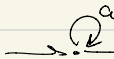
Зад: Да се построи к.св.ур. с език $\mathcal{L} = \{a^n b^m c^k \mid k \le \max(n,m)\}$

$k > \min(m,n)$ $\quad$ $k \le \max(m,n)$

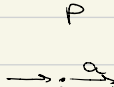
$k > n$ или $k > m$ $\quad$ $k \le n$ (или) $k \le m$

$\Rightarrow \mathcal{L} = \{a^n b^m c^k \mid k \le n \text{ или } k \le m\} = \{a^n b^m c^k \mid \underbrace{k \le m}_{k+\ell+1=m}, \ell \in \mathbb{N}\} \cup \{a^n b^m c^k \mid \underbrace{k \le n}_{k+p+1=n}, p \in \mathbb{N}\}$

$= \{a^n b^{\ell+1} b^k c^k \mid k, \ell, m \in \mathbb{N}\} \cup \{a^{p+1} a^k b^m c^k \mid k, p, m \in \mathbb{N}\} = \underbrace{\{a\}^* \{b\}^+ \{b^k c^k \mid k \in \mathbb{N}\}}_{\mathcal{L}_1} \cup \underbrace{\{a^k w c^k \mid k \in \mathbb{N}, w \in a^+ b^+\}}_{\mathcal{L}_2}$



$a^* b^+$ $\quad G_{a^*b^+} = \langle \{a,b,c\}, \{p,q\}, \{p \to aq \mid bq, q \to bq \mid \varepsilon\}, p \rangle$



$a^+ b^*$ $\quad G_{a^+b^*} = \langle \{a,b,c\}, \{s,t,r\}, \{s \to at, t \to at \mid br \mid \varepsilon, r \to br \mid \varepsilon\}, s \rangle$

$G_2 = \langle \{a,b,c\}, \{T_2, s, t, r\}, \{T_2 \to s \mid aT_2c, s \to at, t \to at \mid br \mid \varepsilon, r \to br \mid \varepsilon\}, T_2 \rangle$

вкючване $a^+b^* \cup a^k c^k$

$G_1 = \langle \{a, b, c\}, \{T_1, S_1, p, q\}, \{T_1 \to pS_1, S_1 \to bS_1c \mid \varepsilon, p \to ap \mid bq, q \to bq \mid \varepsilon\}, T_1 \rangle$

$G_2 = \langle \{a, b, c\}, \{S, T_1, S_1, p, q, T_2, s, t, v\}, \{S \to T_1 \mid T_2\} \cup P_1 \cup P_2, S \rangle$